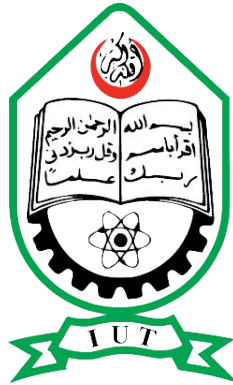


# ISLAMIC UNIVERSITY OF TECHNOLOGY



MACHINE LEARNING LAB

CSE 4622

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## Updated Term Project Final Guidelines

*A Comprehensive Guide for Students*

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*Course:*

CSE 4622 – Machine Learning Lab

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Computer Science & Engineering  
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# 1 Final Presentation

## 1.1 Overview

The final presentation showcases your completed work: what you set out to do, what you accomplished, your results, and your insights. It is worth **20%** of the total project grade.

The final presentation is **mandatory**. Absence without prior approval results in a zero for this component.

## 1.2 Format and Duration

- **Duration:** 8 minutes presentation + 3–4 minutes Q&A.
- **Slides:** Required, submitted as PDF by 11:59 PM the night before your presentation.
- **Demo (optional):** Encouraged if your project has a tool/interactive component; must fit the 8-minute slot.
- **Participation:** Every team member must speak, or they receive no credit for this component.

## 1.3 Recommended Slide Structure

**Topic 1. Title Slide** – project title, team name/members, date.

**Topic 2. Problem and Motivation** – 1–2 slides.

**Topic 3. Related Work** – what was done before, what's new here.

**Topic 4. Dataset Description** – source, size, samples, class distribution, challenges.

**Topic 5. Methodology** – algorithms, pipeline diagram, preprocessing, evaluation strategy.

**Topic 6. Experiments and Results** – tables/figures/confusion matrices/learning curves; multiple metrics, not just accuracy.

**Topic 7. Analysis and Discussion** – what results mean, where/why models fail, surprises.

**Topic 8. Limitations** – honest acknowledgment of shortcomings.

**Topic 9. Conclusion and Future Work.**

**Topic 10. References.**

## 1.4 Final Presentation Grading Rubric

### 1.4.1 Team-wise Grading

| Criterion                                                  | Description                                                                                                                                            | Marks |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Research Question, Motivation, and Background Study</b> | Problem clearly restated? Motivation compelling? Scope well-defined? Grounding in prior work?                                                          | 10    |
| <b>Dataset, Preprocessing, and Methodology</b>             | Final dataset described in detail? Preprocessing documented/justified? Data visualizations shown? Models used?                                         | 20    |
| <b>Results and Analysis</b>                                | Multiple metrics used? Hyperparameters justified? Baseline comparisons shown? Tables/figures clear and labelled? <i>Why</i> results are what they are? | 20    |
| <b>Slide Quality and Uniqueness of the Project</b>         | Slides visually professional, organized, logical flow, readable figures, appropriate density? Project original/creative?                               | 10    |

### 1.4.2 Individual Grading

| Criterion                              | Description                                                                                          | Marks |
|----------------------------------------|------------------------------------------------------------------------------------------------------|-------|
| <b>Oral Delivery and Participation</b> | Does every member contribute? Clear delivery, well-paced, within time?                               | 10    |
| <b>Response to Q&amp;A</b>             | Accurate, thoughtful answers? Understanding of limitations? Honest “we don’t know” when appropriate? | 10    |

**Tips for a strong final presentation:**

- Lead with results, not process.
- Discuss every figure/table verbally – don't just flash it.
- Prepare for: “*Why this algorithm over alternatives?*” and “*What would you do differently?*”
- Explain numbers, don't just report them – this is what separates strong analysis from average.

## 2 Term Paper

### 2.1 Overview

Every team submits a term paper in the format of a research publication – not a lab report or user manual – following the style of ACL, EMNLP, IEEE, or Springer papers. Worth **20%** of the total grade.

The term paper must be written in  $\text{\LaTeX}$  using the provided IEEE template. Word, Google Docs, or other formats will **not** be accepted.

### 2.2 Length and Format

- **Length:** 6 pages (including references).
- **Template:** [IEEE template link](#).
- **References:** BibTeX for all references; every cited work must appear in the list and vice versa.
- **Figures/Tables:** All must be numbered, titled, and referenced in the text.
- **Submission:** Compiled `.pdf` + full `.tex` source (zipped) + Turnitin **plagiarism report** + **AI writing report**.

### 2.3 Paper Structure

#### 2.3.1 Title

Concise (10–15 words), descriptive, conveys the key contribution.

**Example**

Good: “*PoemSum: A Novel Dataset and Multi-Model Benchmark for Automatic Poem Summarization*”

Good: *“Comparative Analysis of Classical Classifiers for Bangla Handwritten Numeral Recognition Under Noisy Conditions”*

Bad: *“Our Machine Learning Project”*

### 2.3.2 Abstract

Self-contained, **150–250 words**, written last. Must cover: the problem & its significance, the approach, key results, and the main conclusion/contribution. Every sentence must add information – avoid vague statements like “results were satisfactory.”

### 2.3.3 Introduction

**1–1.5 pages**. Must: (1) hook the reader with the broad problem; (2) narrow to your specific problem; (3) state the gap in existing work; (4) explicitly list contributions as bullets (e.g. *“We introduce X...”*, *“We benchmark Z algorithms...”*, *“We show A outperforms B by 5% on M...”*); (5) end with one sentence on paper organization.

### 2.3.4 Related Work

**0.5–1.5 pages**, covering: works on the same/similar problem, datasets used for similar tasks, similar methods, and how your work differs from each group.

Organize thematically (e.g. “Feature-based approaches,” “Deep learning approaches”) with synthesizing prose – do not just list papers. Cite **at least 10** proper academic sources (ACL, EMNLP, CVPR, IEEE, Springer, arXiv with caution). No Wikipedia, blogs, or YouTube.

### 2.3.5 Dataset

Include: source, size/format and train/val/test split, class distribution (chart/table), feature and target description, representative samples, preprocessing steps with justification, and known limitations/biases. If self-collected, describe annotation process, annotator agreement, and quality verification.

### 2.3.6 Methodology

**1–2.5 pages**. Include: formal problem definition; for each algorithm – brief description, key hyperparameters and how chosen (grid search/CV), any modifications; feature engineering rationale; experimental setup (split ratios, CV folds, random seeds for reproducibility); every evaluation metric defined and justified (not just “accuracy”); baselines (at minimum a majority-class or simple baseline).

### 2.3.7 Results

Include: a main results table comparing all models across all metrics on the test set; confusion matrices for the best model (classification tasks); learning/convergence curves if applicable; ablation studies if applicable.

Report results objectively – do not cherry-pick metrics.

### 2.3.8 Discussion

Interpret the results: what they mean for your research question; why some models outperform others (data, features, or algorithm assumptions); the hardest cases and common errors; consistency with literature; what you'd do differently with more time/resources.

### 2.3.9 Conclusion

**1–2 paragraphs.** Summarize contributions/findings, restate what was learned, point to future work. No new information, and no reflective statements like “this was a valuable learning experience.”

### 2.3.10 References

Consistent BibTeX formatting via `natbib`. Include for papers: authors, title, venue, year, volume/pages; for datasets: authors/org, name, year, URL; for software: package name, version, URL. **At least 10** references, all from legitimate academic sources.

## 2.4 Writing Guidelines

1. Write in third person/passive voice (“the results suggest” not “we think”).
2. Be precise and concise – eliminate filler.
3. Define all notation on first use.
4. Number all equations, figures, and tables; refer to them by number.
5. Avoid informal language (“did pretty well” → “achieved an F1-score of 0.87”).
6. Proofread carefully – grammar/spelling errors are unacceptable.
7. No plagiarism, including self-plagiarism or copying from other teams – referred to the disciplinary committee.
8. Do not have AI write the whole paper – severe marks deduction.

## 2.5 Term Paper Grading Rubric

| Criterion                             | Description                                                                                                                                               | Marks |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Abstract and Introduction</b>      | Self-contained abstract? Well-motivated intro? Explicit contributions?                                                                                    | 10    |
| <b>Related Work</b>                   | $\geq 10$ relevant papers cited/discussed? Organized thematically? Novelty distinguished? Citations correct?                                              | 10    |
| <b>Dataset Description</b>            | Well-documented (size, class distribution, features, preprocessing)? Samples shown? Limitations acknowledged?                                             | 10    |
| <b>Methodology</b>                    | Problem formally defined? Algorithms described in sufficient detail? Rigorous experimental setup (splits, metrics, baselines)? Hyperparameters justified? | 15    |
| <b>Results</b>                        | Complete, honestly reported results? Multi-metric, multi-model comparisons? Figures/tables well-labelled and informative?                                 | 15    |
| <b>Discussion and Analysis</b>        | Interpretation beyond raw numbers? Failure cases analyzed? Insight into why algorithms succeed/fail?                                                      | 15    |
| <b>Originality of Contribution</b>    | Original contribution? Non-trivial research question?                                                                                                     | 10    |
| <b>Writing Quality and Formatting</b> | Well-written, proofread, consistently formatted? All sections present? Citations complete/consistent?                                                     | 10    |

| Criterion                                    | Description                                                                                                                 | Marks      |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------|
| Use of $\text{\LaTeX}$ and Proper Formatting | Compiled in $\text{\LaTeX}$ with provided template? Equations typeset correctly? Figures as vector graphics where possible? | 5          |
| <b>Total</b>                                 |                                                                                                                             | <b>100</b> |

### 3 Summary of Grading

| Component             | Weight     | Continuous Evaluation |
|-----------------------|------------|-----------------------|
| Proposal Presentation | 10%        | —                     |
| Final Presentation    | 20%        | —                     |
| Term Paper            | 20%        | —                     |
| <b>Total</b>          | <b>50%</b> | <b>50%</b>            |

**Late submission policy:** All deadlines are firm. A penalty of **20% per minute** of delay applies to late submissions. Extension requests must be made in writing at least 24 hours before the deadline and are only granted for serious documented circumstances.

## 4 Academic Integrity and Originality

### 4.1 Plagiarism

All work must be original. Plagiarism – copying text, code, figures, or ideas without proper attribution – is a serious violation, including:

- Copying text from papers/websites/textbooks without citation and quotation marks.
- Submitting another team's work as your own.
- Reusing your own prior coursework without disclosure.
- Copying large portions of code from GitHub/Stack Overflow without attribution.

All papers are checked for plagiarism; a minimum **30%** similarity threshold triggers manual review.

## 4.2 Use of Generative AI Tools

Using LLMs (ChatGPT, Claude, Gemini, etc.) to *write* the term paper is **not permitted**. Permitted uses: proofreading/grammar correction, understanding concepts (as reference, not a citable source), and debugging code.

Any use of AI tools must be disclosed in a short “Author Contributions” / “AI Usage Statement” paragraph at the end of the paper. **Failure to disclose is treated as academic dishonesty.**

## 4.3 Code Attribution

When using/adapting code from an open-source repository: credit the source in a code comment, mention the source in the paper (dataset/methodology section), and ensure the license permits your use.